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Hierarchical graph representation learning with multi-granularity features for anti-cancer drug response prediction

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***Abstract*— Patients with the same type of cancer often respond differently to identical drug treatments due to unique genomic traits. Accurately predicting a patient's response to drug is crucial in guiding treatment decisions, alleviating patient suffering, and improving cancer prognosis. Current computational methods utilize deep learning models trained on extensive drug screening data to predict anti-cancer drug responses based on features of cell lines and drugs. However, the interaction between cell lines and drugs is a complex biological process involving interactions across various levels, from internal cellular and drug structures to the external interactions among different molecules.To address this complexity, we propose a novel Hierarchical graph representation Learning with Multi-Granularity features (HLMG) algorithm for predicting anti-cancer drug responses. The HLMG algorithm combines features at two granularities: the overall gene expression and pathway substructures of cell lines, and the overall molecular fingerprints and substructures of drugs. Subsequently, it constructs a heterogeneous graph including cell lines, drugs, known cell line-drug responses, and the associations between similar cell lines and similar drugs. Through a graph convolutional network model, the HLMG learns the final cell line and drug representations by aggregating features of their multi-level neighbor in the heterogeneous graph. The multi-level neighbors consist of the node self, directly related drugs/cell lines, and indirectly related similar drugs/cell lines. Finally, a linear correlation coefficient decoder is employed to reconstruct the cell line-drug correlation matrix to predict anti-cancer drug responses. Our model was tested on the Genomics of Drug Sensitivity in Cancer (GDSC) and the Cancer Cell Line Encyclopedia (CCLE) databases. Results indicate that HLMG outperforms other state-of-the-art methods in accurately predicting anti-cancer drug responses.**

***Index Terms*— cancer drug response prediction, multi-level features, hierarchical graph representation learning.**

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Availability: "https://github.com/weiba/HLMG". (Corresponding author: Wei Peng, Jianxin Wang .)

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I. Introduction

CANCER continues to be one of the leading causes of death worldwide, posing a severe threat to human health. The heterogeneity of tumors means that patients with the same cancer type may respond differently to identical treatments [1]. Consequently, accurately predicting how cancer patients will respond to anti-cancer drug treatments is crucial for devising personalized therapeutic strategies, potentially increasing patient survival rates and reducing treatment expenses [2]. Recent efforts in preclinical anti-cancer drug screening, such as the Cancer Cell Line Encyclopedia (CCLE) [3] and the Genomics of Drug Sensitivity in Cancer (GDSC) [4] have experimental data on the responses of thousands of cancer cell lines to various drugs. The availability of these public datasets has empowered researchers to develop effective computational methods for predicting responses to anti-cancer drugs.

Current algorithms for predicting anti-cancer drug responses execute their prognostications by constructing deep-learning models to learn the features of cell lines and drugs [5]. These methodologies typically involve two primary stages: first, learning features specific to cell lines and drugs, and second, combining these features to predict cell line response to drugs. During the feature learning phase, existing methods can be broadly categorized into three types based on the granularity of the extracted features of cell lines and drugs: global features-based methods, similarity features-based methods, and substructure features-based methods. Global features-based methods employ fully connected networks(DeepCDR [6])), convolutional neural networks(GraphCDR [7]), or graph neural networks to learn the attributes of cell lines from comprehensive gene expression data [8], [9] or multi-omics data (such as copy number variations and somatic mutation data [10]–[12]). They extract features from drug molecular fingerprints [13], [14] or molecular graphs [6], [7]. Similarity features-based methods assume that similar cell lines will likely show similar responses to comparable drugs. These approaches calculate the similarity between cell lines as a feature based on gene expression [8] or multi-omics data [10], [15], and the similarity between drugs based on their molecular fingerprints [16]. Substructure features-based methods amalgamate features from internal substructures of cell lines and drugs. For instance, DRPreter [17] divides genes within a cell line into multiple pathways, treating these pathways as substructures of the cell line. It employs a graph attention network to learn the features of each pathway. SubCDR [18] categorizes driver genes into subgroups based on their role (oncogenes, tumor suppressor genes, fusion genes, or genes of unknown function). It uses a CNN model to extract features from these substructures. For drugs, SubCDR employs the BRICS [19] algorithm to dissect drug molecular fingerprints into several functional substructures. It then utilizes a recurrent neural network (RNN) to extract drug substructure features.

After learning the features of cell lines and drugs, computational methods employ various strategies to combine these features for the response prediction task. One category of methods concatenates the learned cell line and drug features and inputs them into classifiers such as fully connected layers(FC) [13], deep forest [20], or convolutional neural networks (CNN) [6] for drug response prediction. Other approaches construct a heterogeneous graph using known responses and employ a graph convolution network (GCN) model that updates representations by aggregating features from neighboring nodes in the graph for drug response prediction [7], [8], [10], [14], [21], [22]. A further category employs models such as the Transformer [17] or graph convolutions [18] to learn interaction features between substructures of cell lines and drugs, and then concatenates these features into a fully connected classifier to predict anti-cancer drug responses.

Existing methods for predicting drug responses often focus on specific features of cell lines and drugs, maintaining a consistent granularity level such as global features, similarity features, or substructure features. Moreover, most existing methods combine these features and emphasize the interactions between associated drugs and cell lines. However, the cellular response to drugs involves complex interactions at multiple molecular levels, both within cell lines and externally with drugs. Genes display varying expression levels within cell lines, and their interaction networks form signalling pathways. These pathways, acting as substructures within the cell lines, collaborate to facilitate cellular functions, thereby presenting features at different granularities, ranging from pathway substructures to overall gene expression levels. Similarly, drugs have unique molecular fingerprints with diverse substructures, displaying features across different granularities, ranging from substructures to the overall molecular fingerprint. Moreover, relationships between drugs and cell lines extend beyond direct associations, forming a multi-tiered network with intricate interactions at various granularities. In this study, we propose a Hierarchical graph representation Learning with Multi-Granularity features (HLMG) algorithm for predicting anti-cancer drug responses. This method initially learns cell line and drug features at two granularities: the substructure and overall levels. Subsequently, it constructs a heterogeneous graph that includes cell lines, drugs, known cell line-drug responses, and similar cell lines and drugs. This algorithm employs a graph convolutional network model to learn the final feature representations of both cell lines and drugs by aggregating features of their multi-level neighbor in the heterogeneous graph. The multi-level neighbors include the entity itself, directly associated drugs or cell lines, and indirectly related similar drugs or cell lines. Finally, we calculate the linear correlation coefficients between the features of cell lines and drugs to reconstruct the cell line-drug association matrix to predict anti-cancer drug responses. Experimental results demonstrate that our model can predict anti-cancer drug responses more accurately than existing methods on the GDSC and CCLE datasets. Specifically, our model excels in response prediction tasks for new cells and new drugs. In the GDSC dataset, our model achieves a 0.49% improvement in AUC values for cell lines and a 1.34% improvement for new drugs compared to the second-best method. Similarly, in the CCLE dataset, our model demonstrates a 0.43% improvement in AUC values for cell lines and a remarkable 6.80% improvement for new drugs over the second-best method. Furthermore, case studies illustrate that HLMG can be a robust tool for predicting new anti-cancer drug responses.

In summary, our model offers the following contributions:

- We propose a Hierarchical graph representation Learning with Multi-Granularity features (HLMG) algorithm for predicting anti-cancer drug responses. This approach considers gene expression levels as the overall features of cell lines and leverages pathways as their substructures. Additionally, it views the molecular fingerprint of a drug as its overall feature and extracts substructures from the drug's molecular fingerprint. Both cell lines and drug features are learned from overall and substructural perspectives.
- We construct a heterogeneous network wherein cell lines and drugs as nodes, by incorporating known cell line-drug responses and associations based on cell line similarities and drug similarities. Within this network, cell lines aggregate multi-granularity features from themselves, directly connected drugs(first-order connections), and similar drugs (second-order connections) to learn their feature representations. Similarly, drugs also aggregate multi-granularity features from themselves, directly connected cell lines(first-order connections), and similar cell lines (second-order connections) to learn their feature representations.
- Extensive experimental results demonstrate that the HLMG algorithm outperforms existing state-of-the-art algorithms across various metrics, indicating its superior performance in predicting anti-cancer drug responses.

II. MATERIALS

In this paper, we address the prediction of anti-cancer drug responses by framing it as a link prediction problem, aiming to predict associations between drugs and cell lines. To evaluate the performance of our model, we conducted experiments on two datasets: GDSC and CCLE. From GDSC, we downloaded

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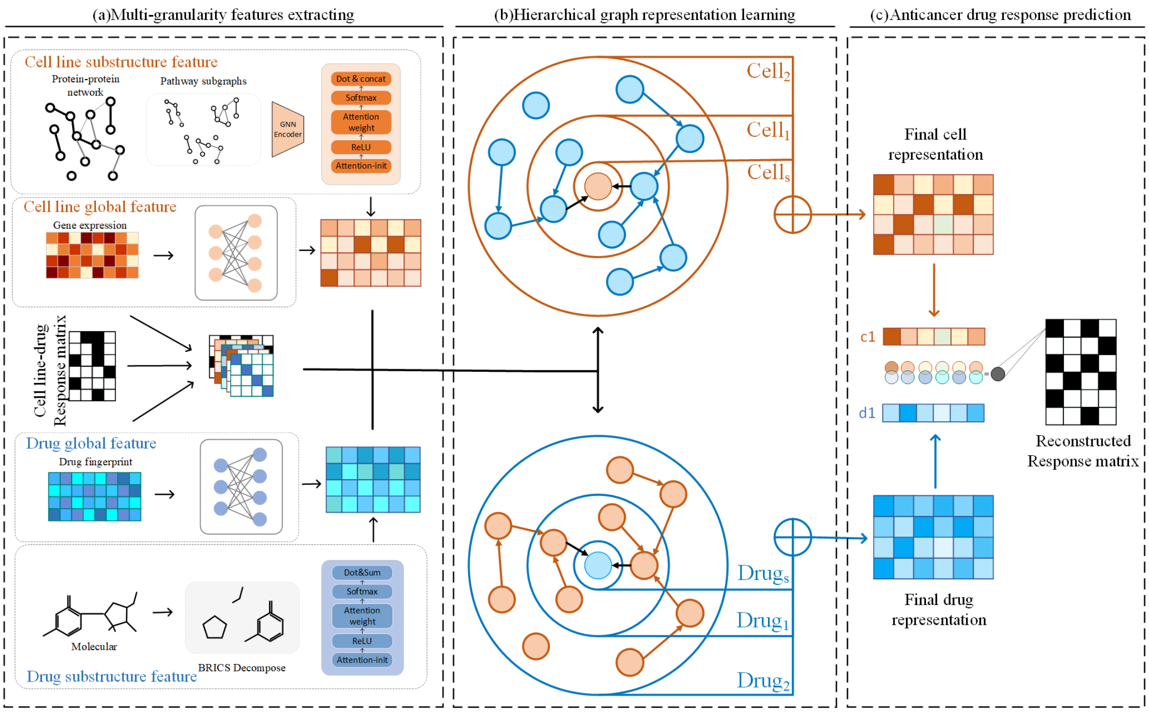

Fig. 1. **The framework of the HLMG.**(a)It extracts cell line and drug features at multi-granularity levels:global-level and substructure features.(b) It inputs the cell line and drug multi-granularity feature into the heterogeneous network and employs hierarchical aggregation module including self-layer, first-order associations layer, and second-order associations layer to learn feature representations of cell line and drug separately. (c) The drug response is predicted by calculating the correlation coefficient between cell line and drug final representation.

Supplementary Tables S4A[1] and S5C[2]. Supplementary Table S4A provides a log-transformed matrix of the maximum half-inhibitory concentration (IC50) values for all screened cell line/drug combinations, encompassing 990 cancer cell lines and 265 test drugs. Supplementary Table S5C lists the sensitivity thresholds for the 265 drugs. A drug is considered sensitive in a cell line if its IC50 value in Supplementary Table S4A does not exceed the threshold in Supplementary Table S5C; otherwise, it is deemed resistant to that cell line. In this paper, the preprocessing of the benchmark dataset for drug response prediction is similar to that described in [14]. After preprocessing, we obtained responses from 962 cell lines to 228 drugs in GDSC, including 20,851 sensitive samples and 156,512 resistant samples. The CCLE database provides records of 11,670 cell line drug assays. Following the previous work [8], [23], we normalized the IC50 values in the dataset. A cell line is considered sensitive to a drug if the normalized IC50 value is below -0.8; otherwise, the cell line is considered resistant to the drug. After preprocessing, we obtained responses from 436 cell lines to 24 drugs, containing 1,696 sensitive samples and 8,768 resistant samples. The methodology involves gene expression and pathway data of cell lines, along with molecular fingerprints of drugs. The drug molecular fingerprints were from the PubChem database [24]. Gene expression data were from the GDSC and CCLE databases, while pathway information was from the KEGG database [25].

[1]`https://www.cancerrxgene.org/gdsc1000/GDSC1000_WebResources//Data/suppData/TableS4A.xlsx`
[2]`https://www.cancerrxgene.org/gdsc1000/GDSC1000_WebResources//Data/suppData/TableS5C.xlsx`

Utilizing the approach in [17], we identified 34 pathways involving 2,369 genes related to cell signaling, cell cycle, and apoptosis. The binary response matrix was constructed based on cell line-drug sensitivity thresholds in the dataset. A '1' indicates sensitivity, while a '0' resistance, with missing values imputed as 0. During model training, missing values were masked and excluded from optimizing the model.

III. METHODS

A. Overview

We devised a hierarchical graph representation learning with multi-granularity features algorithm to predict the cell line responses to anti-cancer drugs. Our method involves three steps to complete the predictive task (see Fig 1). Firstly, we extract cell line and drug features at two granularity levels: overall features and substructure features. Subsequently, we construct a heterogeneous network with nodes representing cell lines and drugs. This network incorporates known cell line-drug responses and associations based on cell line similarity and drug similarity. Within the network, cell lines aggregate multi-granularity features from themselves, directly connected drugs, and second-order connected similar drugs to learn their feature representations. Similarly, drugs integrate multi-granularity features from themselves, directly connected cell lines, and second-order related similar cell lines to learn representations for drugs. Finally, we reconstruct their associations by computing the linear correlation coefficients between cell lines and drugs. This reconstruction facilitates predicting anti-cancer drug responses.

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B. Cell Line Multi-Granularity Feature Extraction

1) Extacting Global-Level Features of Cell lines: We obtained cell lines' global-level features by referring to their gene expression profiles. Initially, we normalize the expression level of each gene within the cell line using Eq 1 to derive C_g:

$$C_{g(i)} = \frac{(c_i - u_i)}{\sigma_i}, \tag{1}$$

where c_i denotes the expression value of the ith gene across all cell lines, with the mean u_i and standard deviation σ_i. The cell line global-level features $H_g \in R^{m\times d_g}$ is calculated by Eq 2:

$$H_g = C_g \cdot W_g, \tag{2}$$

where $C_g \in \mathbb{R}^{m\times d_{cg}}$ is normalized gene expression values of cell lines obtained by Eq 1, m is the number of cell lines, and d_{cg} is the number of genes. $W_g \in \mathbb{R}^{d_{cg}\times d_g}$ is learnable parameters.

2) Extracting Substructure Features of Cell lines: Within cells, the interactions of genes form various pathways that execute specific functions. Each pathway acts as a substructure of the cell. Following the method in [17], we selected 34 cancer-related pathways from the KEGG database [25]. We employ a Graph Convolutional Neural Network model [26] to learn the latent feature vectors of each cell line substructure. Let $P_c = [P_1, P_2, ..., P_{34}]$ be the pathway set. The feature vectors of individual pathway can be computed using Eq 3:

$$\begin{aligned} c_p^i &= GCN^i\left(P_i, p_{gene}^i\right) W^i \\ &= \left(\left(LP_i \cdot p_{gene}^i\right)^T \cdot W_1^i\right) W_2^i, \\ LP_i &= D_{pi}^{-\frac{1}{2}} P_i D_{pi}^{-\frac{1}{2}} \end{aligned} \tag{3}$$

where, $i \in [1, 2, ..., 34]$ denotes the ith pathway. D_{pi} is the degree matrix of the ith pathway. $p_{gene}^i \in \mathbb{R}^{ni\times 1}$ represents the gene expression matrix of the ith pathways. ni indicates the number of genes in the ith pathway. $W_1^i \in \mathbb{R}^{ni\times ni}$ represents the graph convolutional weight parameters for the ith pathway. $W_2^i \in \mathbb{R}^{ni\times d_{pi}}$ is the linear transformation parameter of the ith pathway.

To fully capture the interrelationships between different pathways, we compute the attention weights by Eq 4:

$$V_{ap} = \varphi_{atten}\left(C_p, W\right) = \delta\left(\sigma\left(C_p \cdot W_{ap1}\right) W_{ap2}\right), \tag{4}$$

where σ represents the ReLU activation function. δ denotes Softmax function. $C_p \in \mathbb{R}^{m\times p\times d_{pi}}$ is the pathway feature matrix, $V_{ap} \in \mathbb{R}^{m\times p\times 1}$ is the pathway attention coefficient matrix. m is the number of cell lines, p stands for the number of pathways. d_{pi} is the dimensionality of pathway features. $W_{ap1} \in \mathbb{R}^{d_{pi}\times d_{ca}}$ and $W_{ap2} \in \mathbb{R}^{d_{ca}\times 1}$ are the learnable pathway attention weight parameters.

We combine the pathway feature representations with their respective attention weights and concatenate the representations of each pathway, ultimately yielding the substructure features specific to the cell line, $H_p \in \mathbb{R}^{m\times d_p}$ (see Eq 5)

$$H_p = \left[V_{\text{ap}}^1 \odot c_p^1 \parallel V_{\text{ap}}^2 \odot c_p^2 \parallel \cdots \parallel V_{\text{ap}}^i \odot c_p^i\right], \tag{5}$$

where $\odot$ is the element-wise multiplication, $i \in [1, 2, ..., 34]$ refers to the ith pathway, V_{ap}^i denotes the attention weight for the ith pathway, C_p^i represents the feature matrix of the ith pathway, and $\parallel$ indicates the concatenation operation in this work.

We employ Eq 6 to synthesize the cell line's multi-granularity features, thereby obtaining the cell line's initial feature X_{cell}:

$$X_{cell} = \left[H_g \parallel H_p\right]. \tag{6}$$

C. Drug Multi-Granularity Feature Extraction

The efficiency of drugs is closely associated with the drug's molecular structure and their substructures [27]. Therefore, we consider two levels of granularity features for drugs: global-level and substructure features:

1) Extracting Global-Level Features of Drugs: We perform a linear transformation on the drug binary morgan fingerprint to obtain the drug global-level features $H_f \in \mathbb{R}^{n\times d_f}$:

$$H_f = D_f \cdot W_f, \tag{7}$$

where $D_f \in \mathbb{R}^{n\times d_d}$is the drug molecular fingerprint matrix, n represents the number of drugs, d_d is the length of the drug molecular fingerprint feature and $W_f \in \mathbb{R}^{d_d\times d_f}$is a learnable parameter.

2) Extracting Substructure Features of Drugs: We utilize the BRICS [19] algorithm to break down drug molecules and extract their substructure features based on predetermined chemical rules. This process identifies potential cleavage points within a molecule by targeting chemically labile bonds that align with common drug degradation patterns, such as frequent breaking points in metabolic processes or sites of bond formation in synthesis. Despite the molecular breakdown, the resultant fragments retain the essential chemical information and structural characteristics from the original molecule. We decompose the drug's Smiles into a sequence of substructures and transform these substructures into binary Morgan fingerprint features. We employ Eq 8 to calculate the attention weights for the substructures:

$$V_{ad} = \varphi_{atten}\left(D_s, W\right) = \delta\left(\sigma\left(D_s \cdot W_{ad1}\right) W_{ad2}\right), \tag{8}$$

where σ and δ are the ReLU and Softmax function, respectively. $D_s \in \mathbb{R}^{n\times s\times d_s}$ is the binary Morgan fingerprints of substructures derived from the drug's Smiles via the BRICS algorithm. $V_{ad} \in \mathbb{R}^{n\times s\times 1}$ denotes the substructure attention weight matrix, n is the number of drugs, s is the number of substructures, d_s is the dimension of substructure features. $W_{ad1} \in \mathbb{R}^{d_s\times d_{da}}$ and $W_{ad2} \in \mathbb{R}^{d_{da}\times 1}$ are parameters for the substructure attention weights.

We sum up all substructure features by Eq 9 to produce the drug substructure feature representation, $H_s \in \mathbb{R}^{n\times d_s}$:

$$H_s = sum\left(V_{ad} \odot D_s\right). \tag{9}$$

After that, we employ Eq 10 to integrate the drug's multi-granularity feature representation, denoted as X_{drug}:

$$X_{drug} = \left[H_f \parallel H_s\right]. \tag{10}$$

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D. Aggregating Multi-Level External Features for Cell lines

Based on the hypothesis that similar cell lines have similar reactions to similar drugs, we establish a heterogeneous network where nodes represent cell lines and drugs. This network encompasses known interactions between cell lines and drugs and associations derived from cell line similarity and drug similarity. Cell lines learn their final feature representations by aggregating multi-level features from themselves, directly connected drugs, and similar drugs (second-order connections) within this heterogeneous network. Let $A \in \{0,1\}^{(m\times n)}$ store the known response relationships between cell lines and drugs, where the rows and columns correspond to cell lines and drugs. A value of 1 in A_{cd} indicates that cell line c has a reaction to drug d; otherwise a value of 0 indicates no reaction. $D_{c(ij)} = \sum_j A_{ij} + 1$ and $D_{d(ij)} = \sum_i A_{ji} + 1$ are the degree matrices of matrix A. We obtain the cell line themselves features, denoted as $Cell_s^{(k)}$, by normalizing their multi-granularity features.

$$\begin{aligned} Cell_s^{(k)} =& Cell_s w_{cs}^{(k-1)} + Cell_s \odot Cell_s^0 w_{cs}^{(k-1)}, \\ Cell_s =& \left(D_c^{-1} + I_c\right) Cell_s^{(k-1)}. \end{aligned} \tag{11}$$

The terms $cell_s^0$ and $Drug_s^0$ represent the multi-granularity features of cell lines and drugs, obtained from Eq 6 and Eq 10, respectively. The k denotes the number of aggregation iterations. Here, k is set to 1. I_c is an (m*m) identity matrix.

We get the first-order external features of the cell lines, denoted as $cell_1^{(k)}$, by aggregating the features of the drugs directly connected in the heterogeneous network (see Eq 12). Here, $Drug_s^{(k-1)}$ represents the drug self features of the $(k-1)\,th$ iteration, as defined in Eq 18. Inspired by [14], in addition to aggregating the nodal features of neighbors, we also consider the interaction between the element-level features of the cell itself and its neighbors (feature-wise multiplication).

$$\begin{aligned} Cell_1^{(k)} =& Cell_1 W_{c1}^{(k-1)} + Cell_1 \odot Cell_s^0 W_{c1}^{(k-1)}, \\ Cell_1 =& \left(D_c^{\frac{1}{2}} A D_d^{\frac{1}{2}}\right) Drug_s^{(k-1)}. \end{aligned} \tag{12}$$

Considering similar drugs have similar responses in similar cell lines, we aggregate the features of similar neighboring drugs to learn the second-order external feature of the cell lines. Let matrix B store the similarity relationships between drugs. Let T_i denote the top T drugs most similar to drug i. Here, T is set to 7. Thus, the drug similarity matrix B is defined as follows:

$$B_{ij} = \begin{cases} S_{d(ij)} & j \in T_i \\ 0 & j \notin T_i \end{cases} \tag{13}$$

$$S_{d(ij)} = \left(\frac{|d_i \cap d_j|}{|d_i \cup d_j|} + e^{-\frac{\|A_i^T - A_j^T\|^2}{2\varepsilon^2}}\right)/2, \tag{14}$$

where A^T represents the transpose of the known response matrix of cell lines to drugs and d_i is the molecular fingerprint features of the ith drug.

We obtain the second-order external features of the cells by collecting the features of drugs that are similar to those directly linked with the cell lines. The second-order external features of the cells is defined as follows:

$$\begin{aligned} Cell_2^{(k)} =& \left(D_c^{\frac{1}{2}} A D_d^{\frac{1}{2}}\right) Drug_{sim}^{(k-1)} W_{c2}^{(k-1)} \\ &+ \left(D_c^{\frac{1}{2}} A D_d^{\frac{1}{2}}\right) Drug_{sim}^{(k-1)} \odot Cell_s^0 W_{c2}^{(k-1)}, \end{aligned} \tag{15}$$

$$Drug_{sim}^{(k-1)} = \sigma\left(\left(D_B^{\frac{1}{2}} B D_B^{\frac{1}{2}}\right) Drug_s^{(k-1)} W_{ds}^{(k)}\right). \tag{16}$$

Wherein, $D_{B(ij)} = \sum_j B_{ij}$ is the diagonal degree matrix of matrix B.

Ultimately, the cell line final representation $Cell_{agg}^{(k)}$ is obtained by integrating their level features:

$$Cell_{agg}^{(k)} = \sigma(Cell_s^{(k)} + Cell_1^{(k)} + Cell_2^{(k)}), \tag{17}$$

where σ denotes the ReLU activation function.

E. Aggregating Multi-Level External Features for Drugs

Similarly, the drug learns its final feature representation by aggregating multilevel features from itself, cell lines directly connected to it, and similar cell lines (second-order connections) within the heterogeneous network. We obtain the drug self features, denoted as $Drug_s^{(k)}$, by the following equation.

$$\begin{aligned} Drug_s^{(k)} =& Drug_s w_{ds}^{(k-1)} + Drug_s \odot Drug_s^0 w_{ds}^{(k-1)}, \\ Drug_s =& \left(D_d^{-1} + I_d\right) Drug_s^{(k-1)}, \end{aligned} \tag{18}$$

where, I_d is an (n*n) identity matrix. Through Eq 19, the drug updates its first-order external features, denoted as $Drug_1^{(k)}$, by aggregating features of cell lines directly connected within the heterogeneous network.

$$\begin{aligned} Drug_1^{(k)} =& Drug_1 W_{d1}^{(k-1)} + Drug_1 \odot Drug_s^0 W_{d1}^{(k-1)}, \\ Drug_1 =& \left(D_d^{\frac{1}{2}} A^T D_c^{\frac{1}{2}}\right) Cell_s^{(k-1)}. \end{aligned} \tag{19}$$

Considering that similar drugs tend to have similar responses in similar cell lines, we sum up the features of similar neighboring cell lines to learn the second-order external feature representation of drugs. Let matrix C store the similarity relations between cell lines. We compute the similarity between cell lines i and j by Eq 21. Let K_i denote the top K cell lines most similar to cell line i. Here, K is set to 7. The cell line similarity matrix C is defined as follows:

$$C_{ij} = \begin{cases} S_{c(ij)} & j \in K_i \\ 0 & j \notin K_i \end{cases} \tag{20}$$

$$S_{c(ij)} = (e^{-\frac{\|x_i - x_j\|_2}{2\varepsilon^2}} + e^{-\frac{\|A_i - A_j\|^2}{2\varepsilon^2}})/2, \tag{21}$$

where X_i represents the normalized gene expression values of cell line i, and A_i corresponds to the connectivity information of cell line i within matrix A. We acquire the drug second-order external features by aggregating the features of cell lines similar to those directly associated with the drug in the heterogeneous network. The formula is defined as follows:

$$\begin{aligned} Drug_2^{(k)} =& \left(D_d^{\frac{1}{2}} A^T D_c^{\frac{1}{2}}\right) \underset{sim}{Cell}^{(k-1)} W_{d2}^{(k-1)} \\ &+ \left(D_d^{\frac{1}{2}} A^T D_c^{\frac{1}{2}}\right) \underset{sim}{Cell}^{(k-1)} \odot Drug_s^0 W_{d2}^{(k-1)}, \end{aligned} \tag{22}$$

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$$Cell_{sim}^{(k-1)} = \sigma\left(\left(D_C^{\frac{1}{2}} C D_C^{\frac{1}{2}}\right) Cell_s^{(k-1)} W_{cs}^{(k)}\right), \tag{23}$$

$D_{C(ij)} = \sum_j C_{ij}$ is the diagonal matrix corresponding to matrix C. Ultimately, we obtain the drug final feature representation, $Drug_{agg}^{(k)}$ by integrating drug three level features:

$$Drug_{agg}^{(k)} = \sigma(Drug_s^{(k)} + Drug_1^{(k)} + Drug_2^{(k)}). \tag{24}$$

F. Anticancer Drug Response Prediction

We adopt Eq 25 to compute the linear correlation coefficient between the cell line's and the drug's final features

$$Corr\left(h_i, h_j\right) = \frac{(h_i-\mu_i)(h_j-\mu_j)^T}{\sqrt{(h_i-\mu_i)(h_i-\mu_i)^T}\sqrt{(h_j-\mu_j)(h_j-\mu_j)^T}}, \tag{25}$$

where $h_i \in Cell_{agg}^{(k)}$ and $h_j \in Drug_{agg}^{(k)}$ denote the features of the ith cell line and the jth drug, respectively. The cell line-drug association matrix is reconstructed as follows:

$$\hat{A} = \varphi\left(Corr\left(Cell_{agg}^{(k)}, Drug_{agg}^{(k)}\right)\right), \tag{26}$$

where $\varphi\left(h\right) = \frac{1}{1+e^{-\gamma h}}$ represents the activation function, and the values in $\hat{A}$ correspond to the predicted anticancer drug response labels. We employ the following loss function to constrain the model:

$$\begin{aligned} L\left(A, \hat{A}\right) = &- \frac{1}{m \times n} \sum_{i,j} M_{ij}[A_{ij} ln\left(\hat{A}_{ij}\right) \\ &+ (1 - A_{ij})\, ln\left(1 - \hat{A}_{ij}\right)]. \end{aligned} \tag{27}$$

The matrix M denotes whether the cell line-drug association is in the training set. If the relationship between cell line i and drug j is for training, then $M_{ij} = 1$, otherwise, $M_{ij} = 0$. Algorithm 1 gives the pseudo-code for HLMG.

Algorithm 1 HLMG method

Input: Gene expression matrix of cell line C_g; pathway features matrix of cell line C_p; pathway subgraphs P_c; drug fingerprint matrix D_f; drug substructure matrix D_s; known cell line-drug association matrix A; sigmoid scaling parameter γ; learning rate lr; number of epochs e.
Output: Reconstructed cell line-drug association matrix.

Calculate the C_g of cell line gene expression matrix by Eq 1
Calculate the matrices I_m, I_n, A, B, C and building the heterogeneous network.
while trained epochs $< e$ **do**
  Calculate the global features of cell line by Eq 2
  Calculate the cell line pathway features by Eq 3
  Calculate the pathway attention weights by Eq 4
  Calculate the substructural features of cell line by Eq 5
  Obtain the cell line's multi-granularity feature Eq 6
  Calculate the global features of drug by Eq 7
  Calculate the drug's substructure attention weights by Eq 8
  Calculate the substructural features of drug by Eq 9
  Obtain the drug's multi-granularity feature Eq 10
  Aggregate cell line feature at different levels through Eq 17
  Aggregate drug features at different levels through Eq 24
  Reconstruct the cell line-drug association matrix $\hat{A}$ by Eq 26
  Calculate the Loss by Eq 27.
  Update model parameters by gradient descent and back propagation.
end while
Return $\hat{A}$

IV. Experiment

A. Baseline

We compared our approach with the following baselines to assess it:

- DeepCDR [6]: Utilizes cell line gene expression and drug molecular graphs as input, predicting drug responses through CNN.
- DeepDSC [13]: Employs cell line gene expression and drug molecular fingerprints as input, extracts cell line features through an autoencoder, and predicts drug responses via a multilayer FC.
- MOFGCN [10]: Computes cell line similarity and drug similarity as input. It operates a GCN model to propagate the similarities of cell lines and drugs for drug response prediction.
- GraphCDR [7]: Takes cell line gene expression and drug molecular graphs as input. It leverages a GCN and a contrastive learning framework to predict drug responses.
- NIHGCN [14]: Uses cell line gene expression and drug molecular fingerprints as input. It applies neighborhood interaction with parallel heterogeneous graph convolution to aggregate features of cell lines and drugs for drug response prediction.
- TSGCNN [8]: Utilizes cell line gene expression and drug molecular fingerprints as input. It predicts drug responses by a dual space graph convolution model.
- SubCDR [18]: Uses gene expression subcomponents and drug substructures as input, aggregates subcomponent information through graph convolution, introduces SVD to decompose the association matrix and predict drug responses.
- DRPreter [17]: Takes cell line pathways and drug molecular graphs as input, and diffuses cell line and drug information through a transformer for drug response prediction.
- DeepTTA [28]: Cell line gene expression and drug substructures are used as inputs and drug substructure features are learned by a transformer for drug response prediction.

The parameters for the baseline methods were adopted as recommended in their original papers.

B. Experimental Setup

We conducted the following experiments to evaluate the performance of our model from various aspects:

- Test 1: Random zeroing tests. Randomly zeroing several known cell line-drug associations to evaluate model's ability to recover them.
- Test 2: Experiments on new cell lines or drugs. Zeroing out parts of rows or columns in the cell line-drug association matrix to investigate the models' capacity to predict response for new cell lines or drugs.
- Test 3: Ablation studies. Conducting ablation studies on different modules of our model to understand their contributions.

- Test 4: Case studies on novel drug responses. Carrying out case studies on our model to assess its capability to detect missing drug responses in cell lines.

C. Experiment on Randomly Zeroing Tests (Test 1)

In Test 1, we randomly selected one-tenth of the positive samples and an equal number of negative samples as an independent test set, while the remaining data served as the cross-validation set. We implemented 5-fold cross-validation on the cross-validation set five times, iterating this process five times. For each iteration, one-fifth of the positive samples and an equal number of randomly selected negative samples formed the validation set, with the remaining samples used for training. Following each training session, we selected the model with the highest AUC value on the validation set to predict the independent test set. This procedure was replicated five times, and the average results were reported.

Table I presents the results of the random zeroing tests for all algorithms on the GDSC and CCLE databases. Our model, HLMG, consistently outperformed all baselines in predicting drug responses in cell lines across both datasets. In the GDSC dataset, our model demonstrated an improvement of 0.22% over the second-ranked methods for both AUC (87.76%) and AUPRC (88.23%). Similarly, on the CCLE dataset, the model has shown increases of 0.40% and 0.41% in AUC (86.44%) and AUPRC (86.64%), respectively, compared to the second-best methods NIHGCN. Comparing DeepCDR and DeepDSC, the models such as MOFGCN, GraphCDR, NIHGCN, TS-GCNN, DRPreter, and HLMG, which focus on known cell line-drug relationships, can learn suitable representations to enhance performance. In contrast to GCN-based and attention-based methods (e.g., NIHGCN, DRPreter), HLMG improved performance, which can be attributed to the aggregation of cell line and drug features at different levels.

To validate the correlation between the output of Eq 25 and the half-maximal inhibitory concentration (IC50) values, we conducted regression validation through random zeroing tests in the GDSC and CCLE datasets. Three regression metrics were used to evaluate the performance of each model: Pearson Correlation Coefficient (PCC), Spearman's Rank Correlation Coefficient (SCC), and Root Mean Square Error (RMSE). Table II reveals that our method has commendable results in regression validation tests compared to baseline methods. Because it controls the highest SCC, PCC value, and second best RMSE value in the CCLE dataset, and achieves the highest SCC value and second best PCC and RMSE value in the GDSC dataset.

D. Experiments on New Cell lines or Drugs (Test 2)

In the cell line-drug association matrix, a row represents a cell line, and a column represents a drug. Test 2 clears several rows or columns of the matrix to assess the algorithm's ability to predict responses for new cell lines or drugs. In each experiment, one-fifth of the rows or columns containing positive samples and an equivalent number of randomly selected negative samples were removed to create the test set. The remaining matrix served as the cross-validation set for a five-fold cross-validation process. In each fold, one-fifth of the positive samples and an equal number of random negative samples from the cross-validation set were set aside as the validation set, while the rest were used for training. The model achieving the best AUC result on the validation set was employed to predict the test set, and this process was repeated until all rows or columns were tested. The average results were then reported.

Table III show the comparative results of nine algorithms on the GDSC and CCLE datasets under this test. Our model maintained the best performance in this test. In the GDSC dataset, our model achieves notable improvements in response prediction tasks for new cells. Specifically, for new cells, our model has the AUC of 77.41% and AUPRC of 78.35%, exhibiting enhancements of 0.49% and 0.46%, respectively, over the second-ranked methods DeepTTA. In the case of response prediction for new drugs, the AUC of 71.42% and AUPRC of 70.24% showcase improvements of 1.34% and 0.61%, respectively, compared to the second-best methods DRPreter. In the CCLE dataset, for new cell lines, our method processes the AUC of 79.02% and AUPRC of 77.60%, indicating substantial improvements of 0.43% and 1.12%, respectively. Additionally, for new drugs, the AUC of 72.97% and AUPRC of 70.96% demonstrate remarkable increases of 6.80% and 6.68%, respectively, over the second-best methods TSGCNN. Even when information regarding cell line-drug associations was limited, our model HLMG still demonstrated superior performance due to its hierarchical and multi-granularity features. It could predict a drug's response to cell lines or identify a cell line's reaction to drugs, even in cases where explicit relationships were unavailable.

To assess our model's ability to identify cell line characteristics related to predisposition to targeted therapies, we focused on compounds targeting CDK4 in the GDSC and CCLE databases. Three agents (PD-0332991, AT-7519, CGP-082996) in GDSC and one (PD-0332991) in CCLE were selected. Using k-means clustering on cell line-drug interaction matrices, we created binary cohorts based on HLMG-learned embeddings or original attributes. Visualization via UMAP showed well-defined clusters (Fig 2), and metrics, including the Silhouette Coefficient (SC) and the Davies-Bouldin Index (DBI) for internal validation and the Normalized Mutual Information (NMI), confirmed the model's capability in differentiation. Table IV provides evidence that the model accurately classifies cell lines into sensitive and resistant cohorts, aligning well with the benchmark dataset.

E. Ablation Study (Test 3)

Our HLMG model is designed to effectively learn cell line and drug features for predicting cell line responses to drugs. Throughout this process, it extracts multi-granularity features from the internal structures of cell lines and drugs, considering self-features, first-order and second-order multi-layer interactions with external molecules. To investigate the performance contribution of different components of our model, we designed the following model variants for comparison under the random zeroing experiment. HLMG-G: It uses pathway features as cell line features; HLMG-P: It

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TABLE I
COMPARISON OF DIFFERENT ALGORITHMS UNDER RANDOMLY ZEROING TEST ON GDSC AND CCLE DATASETS

Algorithm	**GDSC**		**CCLE**	
	AUC	**AUPRC**	**AUC**	**AUPRC**
DeepCDR	$0.8030 \pm 3 \times 10^{-3}$	$0.8064 \pm 3 \times 10^{-3}$	$0.8384 \pm 6 \times 10^{-5}$	$0.8292 \pm 2 \times 10^{-4}$
DeepDSC	$0.8083 \pm 7 \times 10^{-5}$	$0.8130 \pm 8 \times 10^{-5}$	$0.8543 \pm 5 \times 10^{-5}$	$0.8489 \pm 5 \times 10^{-5}$
MOFGCN	$0.8685 \pm 3 \times 10^{-6}$	$0.8726 \pm 3 \times 10^{-6}$	$0.8497 \pm 5 \times 10^{-5}$	$0.8514 \pm 7 \times 10^{-5}$
GraphCDR	$0.8176 \pm 6 \times 10^{-5}$	$0.8253 \pm 5 \times 10^{-5}$	$0.8396 \pm 1 \times 10^{-4}$	$0.8367 \pm 1 \times 10^{-4}$
NIHGCN	$0.8754 \pm 3 \times 10^{-6}$	$0.8801 \pm 3 \times 10^{-6}$	$0.8604 \pm 7 \times 10^{-5}$	$0.8623 \pm 6 \times 10^{-5}$
TSGCNN	$0.8706 \pm 3 \times 10^{-6}$	$0.8745 \pm 4 \times 10^{-6}$	$0.8524 \pm 7 \times 10^{-5}$	$0.8542 \pm 6 \times 10^{-5}$
SubCDR	$0.8251 \pm 3 \times 10^{-5}$	$0.8292 \pm 2 \times 10^{-5}$	$0.7802 \pm 7 \times 10^{-5}$	$0.7830 \pm 2 \times 10^{-4}$
DRPreter	$0.8409 \pm 3 \times 10^{-5}$	$0.8473 \pm 3 \times 10^{-5}$	$0.8237 \pm 6 \times 10^{-4}$	$0.8087 \pm 2 \times 10^{-3}$
DeepTTA	$0.8455 \pm 7 \times 10^{-6}$	$0.8494 \pm 7 \times 10^{-6}$	$0.8301 \pm 3 \times 10^{-5}$	$0.8299 \pm 6 \times 10^{-5}$
HLMG	**0.8776** $\pm 2 \times 10^{-6}$	**0.8823** $\pm 3 \times 10^{-6}$	**0.8644** $\pm 6 \times 10^{-5}$	**0.8664** $\pm 9 \times 10^{-5}$

TABLE II
REGRESSION EXPERIMENT RESULTS UNDER RANDOMLY ZEROING TEST ON GDSC AND CCLE DATASETS

Algorithm	**GDSC**			**CCLE**		
	PCC	**SCC**	**RMSE**	**PCC**	**SCC**	**RMSE**
DeepCDR	$0.8507 \pm 1 \times 10^{-3}$	$0.8238 \pm 2 \times 10^{-3}$	$1.7175 \pm 5 \times 10^{-2}$	$0.6157 \pm 3 \times 10^{-4}$	$0.6546 \pm 3 \times 10^{-4}$	$3.0789 \pm 5 \times 10^{-3}$
DeepDSC	$0.8616 \pm 3 \times 10^{-5}$	$0.8341 \pm 4 \times 10^{-5}$	$1.5708 \pm 4 \times 10^{-3}$	$0.5993 \pm 7 \times 10^{-5}$	$0.6294 \pm 8 \times 10^{-5}$	$2.9409 \pm 2 \times 10^{-2}$
MOFGCN	$0.9103 \pm 7 \times 10^{-8}$	$0.8983 \pm 3 \times 10^{-7}$	$1.3352 \pm 3 \times 10^{-6}$	$0.4959 \pm 2 \times 10^{-5}$	$0.5846 \pm 5 \times 10^{-5}$	$2.9888 \pm 7 \times 10^{-5}$
GraphCDR	$0.8447 \pm 4 \times 10^{-5}$	$0.8239 \pm 8 \times 10^{-5}$	$1.7230 \pm 1 \times 10^{-3}$	$0.5783 \pm 3 \times 10^{-4}$	$0.6121 \pm 7 \times 10^{-4}$	$3.2174 \pm 7 \times 10^{-3}$
NIHGCN	$0.9266 \pm 2 \times 10^{-7}$	$0.9143 \pm 3 \times 10^{-7}$	$1.2276 \pm 3 \times 10^{-5}$	$0.6569 \pm 7 \times 10^{-5}$	$0.7049 \pm 1 \times 10^{-4}$	$2.7593 \pm 1 \times 10^{-3}$
TSGCNN	$0.9210 \pm 8 \times 10^{-8}$	$0.9081 \pm 2 \times 10^{-7}$	$1.2334 \pm 4 \times 10^{-4}$	$0.5915 \pm 3 \times 10^{-5}$	$0.6681 \pm 1 \times 10^{-4}$	$2.8719 \pm 3 \times 10^{-4}$
SubCDR	$0.9076 \pm 7 \times 10^{-6}$	$0.8887 \pm 1 \times 10^{-5}$	$1.2660 \pm 3 \times 10^{-3}$	$0.6697 \pm 6 \times 10^{-5}$	$0.6965 \pm 2 \times 10^{-4}$	**2.6498** $\pm 4 \times 10^{-3}$
DRPreter	**0.9309** $\pm 1 \times 10^{-5}$	$0.9113 \pm 2 \times 10^{-5}$	**1.1382** $\pm 1 \times 10^{-3}$	$0.6667 \pm 1 \times 10^{-3}$	$0.6832 \pm 1 \times 10^{-3}$	$2.7939 \pm 3 \times 10^{-2}$
DeepTTA	$0.9136 \pm 2 \times 10^{-6}$	$0.8894 \pm 2 \times 10^{-6}$	$1.2147 \pm 1 \times 10^{-4}$	$0.5776 \pm 8 \times 10^{-5}$	$0.5872 \pm 2 \times 10^{-4}$	$2.8774 \pm 4 \times 10^{-3}$
HLMG	$0.9294 \pm 2 \times 10^{-7}$	**0.9160** $\pm 3 \times 10^{-7}$	$1.2139 \pm 3 \times 10^{-4}$	**0.6753** $\pm 4 \times 10^{-5}$	**0.7213** $\pm 9 \times 10^{-5}$	$2.6840 \pm 9 \times 10^{-4}$

TABLE III
RESPONSE PREDICTION FOR NEW DRUGS OR NEW CELL LINES ON GDSC AND CCLE DATASETS

Algorithm	**GDSC new cell lines**		**GDSC new drugs**		**CCLE new cell lines**		**CCLE new drugs**	
	AUC	**AUPRC**	**AUC**	**AUPRC**	**AUC**	**AUPRC**	**AUC**	**AUPRC**
DeepCDR	$0.6918 \pm 4 \times 10^{-4}$	$0.7008 \pm 5 \times 10^{-4}$	$0.6221 \pm 7 \times 10^{-3}$	$0.6210 \pm 7 \times 10^{-3}$	$0.6736 \pm 8 \times 10^{-3}$	$0.6467 \pm 6 \times 10^{-3}$	$0.6542 \pm 1 \times 10^{-2}$	$0.6402 \pm 8 \times 10^{-3}$
DeepDSC	$0.6592 \pm 8 \times 10^{-3}$	$0.6640 \pm 9 \times 10^{-3}$	$0.6421 \pm 7 \times 10^{-3}$	$0.6476 \pm 8 \times 10^{-3}$	$0.6108 \pm 2 \times 10^{-2}$	$0.6074 \pm 2 \times 10^{-2}$	$0.5745 \pm 1 \times 10^{-2}$	$0.5682 \pm 9 \times 10^{-3}$
MOFGCN	$0.5852 \pm 1 \times 10^{-4}$	$0.5969 \pm 1 \times 10^{-4}$	$0.6845 \pm 3 \times 10^{-4}$	$0.6817 \pm 3 \times 10^{-4}$	$0.6712 \pm 5 \times 10^{-4}$	$0.6230 \pm 5 \times 10^{-4}$	$0.6526 \pm 1 \times 10^{-3}$	$0.6423 \pm 1 \times 10^{-3}$
GraphCDR	$0.6946 \pm 3 \times 10^{-4}$	$0.6792 \pm 3 \times 10^{-4}$	$0.6494 \pm 3 \times 10^{-4}$	$0.6003 \pm 2 \times 10^{-4}$	$0.6330 \pm 7 \times 10^{-4}$	$0.6088 \pm 1 \times 10^{-3}$	$0.5731 \pm 3 \times 10^{-3}$	$0.5465 \pm 1 \times 10^{-3}$
NIHGCN	$0.7689 \pm 3 \times 10^{-5}$	$0.7721 \pm 9 \times 10^{-5}$	$0.6861 \pm 8 \times 10^{-5}$	$0.6709 \pm 3 \times 10^{-4}$	$0.7663 \pm 3 \times 10^{-4}$	$0.7551 \pm 3 \times 10^{-4}$	$0.6381 \pm 1 \times 10^{-2}$	$0.6298 \pm 9 \times 10^{-3}$
TSGCNN	$0.6023 \pm 1 \times 10^{-4}$	$0.6196 \pm 2 \times 10^{-4}$	$0.6593 \pm 5 \times 10^{-4}$	$0.6600 \pm 5 \times 10^{-4}$	$0.7030 \pm 4 \times 10^{-4}$	$0.6547 \pm 6 \times 10^{-4}$	$0.6617 \pm 3 \times 10^{-3}$	$0.6428 \pm 2 \times 10^{-3}$
SubCDR	$0.5964 \pm 1 \times 10^{-4}$	$0.6179 \pm 1 \times 10^{-4}$	$0.6536 \pm 7 \times 10^{-4}$	$0.6536 \pm 3 \times 10^{-4}$	$0.6440 \pm 1 \times 10^{-3}$	$0.6199 \pm 6 \times 10^{-4}$	$0.6162 \pm 2 \times 10^{-3}$	$0.6120 \pm 1 \times 10^{-3}$
DRPreter	$0.7487 \pm 2 \times 10^{-4}$	$0.7631 \pm 1 \times 10^{-4}$	$0.7008 \pm 4 \times 10^{-4}$	$0.6963 \pm 4 \times 10^{-4}$	$0.7586 \pm 8 \times 10^{-4}$	$0.7327 \pm 1 \times 10^{-3}$	$0.6484 \pm 5 \times 10^{-3}$	$0.6291 \pm 4 \times 10^{-3}$
DeepTTA	$0.7692 \pm 8 \times 10^{-5}$	$0.7789 \pm 6 \times 10^{-5}$	$0.6956 \pm 2 \times 10^{-4}$	$0.6872 \pm 2 \times 10^{-4}$	$0.7859 \pm 3 \times 10^{-4}$	$0.7648 \pm 4 \times 10^{-4}$	$0.6566 \pm 4 \times 10^{-3}$	$0.6349 \pm 3 \times 10^{-3}$
HLMG	**0.7741** $\pm 2 \times 10^{-4}$	**0.7835** $\pm 3 \times 10^{-4}$	**0.7142** $\pm 3 \times 10^{-4}$	**0.7024** $\pm 3 \times 10^{-4}$	**0.7902** $\pm 2 \times 10^{-4}$	**0.7760** $\pm 4 \times 10^{-4}$	**0.7297** $\pm 2 \times 10^{-3}$	**0.7096** $\pm 3 \times 10^{-3}$

employs gene expression features as cell line features; HLMG-F: It uses drug substructure features; HLMG-S: It merely utilizes molecular fingerprint as drug features; HLMG-FA: Its aggregates self and second-order related neighbor features, ignoring first-order associations; HLMG-Self: Its aggregates first-order and second-order related neighbor features, ignoring self features; HLMG-SA: Its aggregates self features and first-order related neighbor features, ignoring second-order related neighbor features; HLMG-Self-SA: Its aggregates first-order related neighbor features, ignoring self features and second-order associations; HLMG-FA-SA: Its aggregates self features, ignoring first-order and second-order associations; HLMG-Self-FA: Its aggregates second-order related neighbor features, ignoring self features and first-order associations; HLMG-residual: It aggregate neighbor features without considering element-level interactions.

From Table V, we observed that the global-level features of cell lines and drugs play a more significant role in predicting anti-cancer drug responses than substructure features. Nevertheless, the substructures serve as a valuable complement to

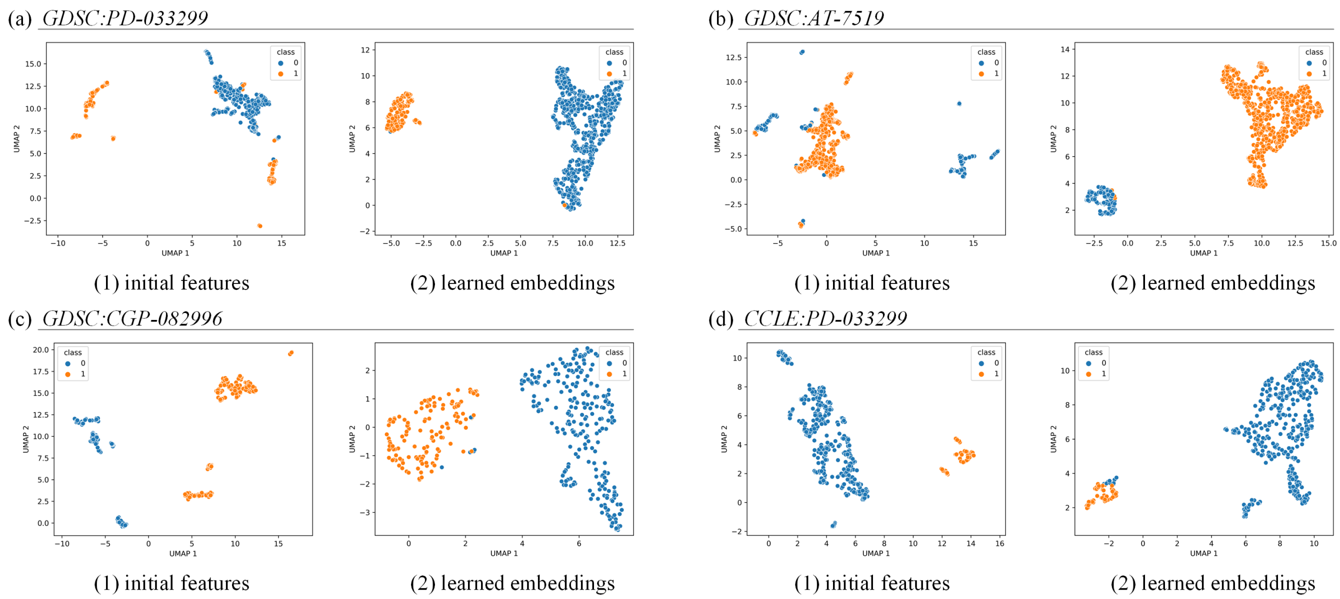

Fig. 2. **UMAP visualization of cell line clustering based on original features or learned embeddings**

TABLE IV
RESULTS OF CELL LINE CLUSTERING BASED ON ORIGINAL FEATURES OR LEARNED EMBEDDINGS

Datasets	Target drugs		Silhouette	DBI	NMI
GDSC	PD-0332991	Initial features	0.4996	1.4555	0.0257
		Learned embeddings	**0.7365**	**0.3082**	**0.0262**
	AT-7519	Initial features	0.4122	2.4428	0.1205
		Learned embeddings	**0.7508**	**0.2571**	**0.2487**
	CGP-082996	Initial features	0.5845	0.6731	0.0314
		Learned embeddings	**0.5981**	**0.5671**	**0.0347**
CCLE	PD-0332991	Initial features	0.6263	0.4106	0.0185
		Learned embeddings	**0.6635**	**0.3115**	**0.0194**

the features of cell lines and drugs, consequently enhancing the overall predictive performance of the model. When updating node features by aggregating connective neighborhood features in the heterogeneous network, it is noteworthy that the node self-feature is the most influential, followed by that from the first-order neighbors in the prediction task. The introduction of second-order associative aggregation has little impact on the outcome. Ultimately, our model achieved optimal results by synthesizing information from these three distinct hierarchical levels.

F. Case Analysis (Test 4)

Previous studies uncovered that approximately 20% of responses are missing in the GDSC dataset [7]. To further test our model's capacity to detect novel drug responses within cell lines, we trained it using all known cell line drug responses from the GDSC dataset, subsequently predicting unknown responses within the GDSC data. We evaluated two clinically approved drugs, GSK690693 and Bortezomib. GSK690693 is an inhibitor targeting the Akt kinase—a crucial intracellular signaling molecule that regulates cell survival, proliferation, metabolism, and apoptosis. Bortezomib is a proteasome inhibitor used to treat multiple myeloma and certain types of lymphoma. Table VI presents the top ten cell lines sensitive to GSK690693 and Bortezomib as predicted by our model. The drug GSK690693 is sensitive to cell lines such as RCH-ACV, MOLT-16, and JEKO-1. Study [29] examined the response of pre-B cell line RCH-ACV and T cell line MOLT-16 to the pan-Akt kinase inhibitor GSK690693, finding that it effectively inhibited proliferation in both RCH-ACV and MOLT-16. Research [30] found that GSK690693 could effectively inhibit the proliferation of the MCL cell line JEKO-1. Regarding Bortezomib, our model predicted that it is sensitive to cell lines A375, AGS, HOS, MEL-JUSO, and MIA-PaCa-2. Reference [31] indicated that Bortezomib possesses inhibitory activity against melanoma cells A375. [32]highlighted that Bortezomib could downregulate the degradation of relevant substrates in AGS cells, promoting apoptosis. Findings from [33] suggest that Bortezomib induces growth inhibition in HOS cells in a time- and dose-dependent manner and triggers autophagy and apoptosis dose-dependently. The autophagy and apoptosis in Bortezomib-responsive HOS cells correspond with changes in the intracellular MAPK signaling molecules' levels. Reference [34] noted that the expression of BCL-2 is high in MIA-PaCa-2, and mutations at the NF-κB site reduce the activity of the BCL-2 promoter in the cell line. Inhibiting NF-κB activity can decrease BCL-2 protein levels, and the apoptotic efficacy of Bortezomib in inhibiting the proteasome is commensurate with its ability to inhibit NF-κB activity and reduce BCL-2 levels.

To further explore the cell line reaction mechanism to drugs, we ranked the substructural attention scores of GSK690693 and Bortezomib, which were calculated using Eq 8. Table VII shows that the substructure '*n1c(*)nc2c(C#CC(C)(C)O)ncc(*)c21' of GSK690693 includes a heterocyclic system, whereas '*c1nonc1N' represents a nitrogen-containing

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TABLE V
ABLATION EXPERIMENTS ON GDSC AND CCLE DATASETS

Algorithm	GDSC		CCLE	
	AUC	AUPRC	AUC	AUPRC
HLMG	$\mathbf{0.8770} \pm 2 \times 10^{-6}$	$\mathbf{0.8814} \pm 2 \times 10^{-6}$	$\mathbf{0.8691} \pm 7 \times 10^{-5}$	$\mathbf{0.8663} \pm 1 \times 10^{-4}$
HLMG-FA	$0.8760 \pm 2 \times 10^{-6}$	$0.8803 \pm 2 \times 10^{-6}$	$0.8658 \pm 8 \times 10^{-5}$	$0.8640 \pm 9 \times 10^{-5}$
HLMG-Self	$0.8726 \pm 2 \times 10^{-6}$	$0.8770 \pm 2 \times 10^{-6}$	$0.8518 \pm 9 \times 10^{-5}$	$0.8517 \pm 1 \times 10^{-4}$
HLMG-SA	$0.8760 \pm 1 \times 10^{-6}$	$0.8801 \pm 2 \times 10^{-6}$	$0.8662 \pm 7 \times 10^{-5}$	$0.8638 \pm 1 \times 10^{-4}$
HLMG-Self-SA	$0.8726 \pm 2 \times 10^{-6}$	$0.8766 \pm 2 \times 10^{-6}$	$0.8515 \pm 1 \times 10^{-4}$	$0.8514 \pm 2 \times 10^{-4}$
HLMG-FA-SA	$0.8755 \pm 1 \times 10^{-6}$	$0.8801 \pm 1 \times 10^{-6}$	$0.8659 \pm 6 \times 10^{-5}$	$0.8622 \pm 1 \times 10^{-4}$
HLMG-Self-FA	$0.8647 \pm 2 \times 10^{-6}$	$0.8703 \pm 3 \times 10^{-6}$	$0.8538 \pm 8 \times 10^{-5}$	$0.8507 \pm 2 \times 10^{-4}$
HLMG-Residual	$0.8740 \pm 3 \times 10^{-6}$	$0.8785 \pm 4 \times 10^{-6}$	$0.8651 \pm 1 \times 10^{-4}$	$0.8642 \pm 2 \times 10^{-4}$
HLMG-G	$0.8686 \pm 7 \times 10^{-7}$	$0.8734 \pm 8 \times 10^{-7}$	$0.8424 \pm 2 \times 10^{-4}$	$0.8427 \pm 4 \times 10^{-4}$
HLMG-P	$0.8758 \pm 2 \times 10^{-6}$	$0.8797 \pm 2 \times 10^{-6}$	$0.8640 \pm 8 \times 10^{-5}$	$0.8616 \pm 2 \times 10^{-4}$
HLMG-F	$0.8749 \pm 5 \times 10^{-7}$	$0.8793 \pm 7 \times 10^{-7}$	$0.8659 \pm 5 \times 10^{-5}$	$0.8634 \pm 3 \times 10^{-5}$
HLMG-S	$0.8757 \pm 5 \times 10^{-8}$	$0.8800 \pm 4 \times 10^{-7}$	$0.8666 \pm 4 \times 10^{-5}$	$0.8631 \pm 1 \times 10^{-4}$

TABLE VI
TOP 10 PREDICTED SENSITIVE CELL LINES FOR GSK690693 AND BORTEZOMIB

Drug	Rank	Cell line	PMID	Drug	Rank	Cell line	PMID
GSK690693	1	CRO-AP2	N/A	Bortezomib	**1**	**A375**	**27677689**
	2	DOHH-2	N/A		**2**	**AGS**	**29843100**
	3	GA-10	N/A		**3**	**HOS**	**23975859**
	4	HGC-27	N/A		4	Hs-633T	N/A
	5	**JEKO-1**	**32120074**		5	KON	N/A
	6	KP-1N	N/A		6	MEL-JUSO	N/A
	7	MOLM-16	N/A		**7**	**MIA-PaCa-2**	**15791457**
	8	**MOLT-16**	**19064730**		8	NCI-H2369	N/A
	9	**RCH-ACV**	**19064730**		9	SNG-M	N/A
	10	SCC90	N/A		10	SW982	N/A

heterocycle, both of which may be crucial for binding to and inhibiting Akt kinase activity. The substructure '*[C@@H](CC(C)C)B (O)O' of Bortezomib contains boron, a key element in Bortezomib, due to its interaction with the catalytic site of the proteasome. We also output and ranked the pathway attention scores for the HOS and AGS cell lines as calculated by Eq 4. Table VIII reports that the HOS cell line is a human osteosarcoma cell line characterized by a high malignant transformation capacity and the ability to form tumors in immunodeficient mice, making it a common model for studying tumor formation, metastasis, and treatment. In drug development, the HOS cell line is also frequently used to assess the efficacy of novel anticancer agents. [35] showed that compounds could increase intracellular reactive oxygen species (ROS) and Ca2+, inducing HOS apoptosis through ROS-mediated mitochondrial dysfunction and inhibition of the mTOR (mammalian target of rapamycin) signaling pathway. Previous literature [33] indicated that Bortezomib induces apoptosis and autophagy in osteosarcoma cells via the mitogen-activated protein kinase (MAPK signaling pathway) in vitro. Research [36] noted that multiple myeloma (MM) is an incurable plasma cell malignancy with aberrant activation of the NF-κB signaling pathway observed in MM, playing a pivotal role in the development of MM by promoting myeloma cell growth, survival, and drug resistance. The AGS cell line is a gastric adenocarcinoma cell line that exhibits epithelial-like morphology in vitro and has a high proliferative capacity. The AGS cell line is of significant value in gastric cancer research, used to study the cellular biological characteristics of gastric cancer, the mechanisms of tumor formation and development, and gene expression in gastric cancer. Additionally, AGS cells are utilized in drug screening and evaluation to find new treatments for gastric cancer. Reference [37] indicates that inhibition of the PI3K/AKT pathway can reduce proliferation induced by proteasome inhibitors and increase apoptosis. Experiments from [38] suggest modulating the NF-κB signaling pathway can induce apoptosis in AGS gastric cancer cells.

These results suggest that our model can detect missing reactions between cell lines and drugs. Moreover, we can find active drug substructures and cell line pathways by assigning higher attention scores.

V. CONCLUSION

In this paper, we developed a hierarchical graph representation learning method to aggregate multi-granularity and multi-level data of drugs and cell lines for the prediction of drug responses. The hierarchical graph representation learning with

TABLE VII
TOP 5 SIGNIFICANT DRUG SUBSTRUCTURES OF CIS AND BORTEZOMIB

Drugs	Rank	Substructure
Cis	1	*n1c(*)nc2c(C#CC(C)(C)O)ncc(*)c21
	2	*c1nonc1N
	3	*O*
	4	*[C@H]1CCCNC1
	5	*CC
Bortezomib	1	*N*
	2	*[C@@H](CC(C)C)B(O)O
	3	*c1cnccn1
	4	*C(*)=O
	5	*C(=O)[C@@H](*)C*

TABLE VIII
TOP 5 SIGNIFICANT PATHWAYS FOR CELL LINE HOS AND AGS

Cell line	Rank	Pathway
HOS	1	mTOR signaling pathway
	2	MAPK signaling pathway
	3	NF-κB signaling pathway
	4	Calcium signaling pathway
	5	Ras signaling pathway
AGS	1	PI3K-Akt signaling pathway
	2	NF-κB signaling pathway
	3	B cell receptor signaling pathway
	4	MAPK signaling pathway
	5	Basal transcription factors

multi-granularity features algorithm constructs representations of cell lines and drugs by integrating multi-granularity features within two internal levels and an external multi-tiered aggregation to predict the responses of cell line-drug pairs. To validate the effectiveness of our model, we compared HLMG with existing state-of-the-art models. The results indicate that the HLMG approach surpasses competing methods. Moreover, we assessed the performance of our model in predicting responses for new cell lines and new drugs through row and column dropout experiments and further substantiated the validity of our model with case studies and ablation experiments, demonstrating the efficacy of using hierarchical aggregation. The experimental outcomes suggest that our model outperforms existing methods in the task of predicting drug responses. However, there is still room to improve our model. In the future, we plan to gather more extensive drug data and explore advanced algorithms to enhance multi-granularity representations of cell lines and drugs.

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COMPETING INTERESTS

The authors declare that they have no competing interests.

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